Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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|---|---|---|---|---|
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## 2. Technology Overview

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| Title of Proposed Technology | Tumor-Targeting Bacteria–Drug Conjugate Platform for Combination Immuno-Oncology Therapy | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1)] | **Technology Field 1 (**Large Category**)** | | **Technology Field 2 (**Large Category**)** | |
| | Bio・Medical | | Bio・Medical | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Medical Bio | Pharmaceutical Bio-based Technologies Diagnostic Devices and Systems | Bio-processes/bio-equipment | Bio-engineering technologies |
| Concept of Proposed Technology | This international joint R&D project aims to develop a **tumor-selective Bacteria–Drug Conjugate (BDC) platform** for anticancer therapy, designed for use in **combination with immune checkpoint inhibitors (ICIs)**. The proposed technology enables localized delivery of highly potent cytotoxic agents directly within the tumor microenvironment by leveraging the intrinsic tumor-homing properties of engineered live bacteria.<br><br>Unlike antibody–drug conjugates (ADCs) or nanoparticle-based delivery systems that rely on passive diffusion or receptor expression, the BDC platform exploits biological features of solid tumors—including hypoxia, necrosis, abnormal vasculature, and local immune suppression—to achieve preferential bacterial accumulation and persistence. This enables access to poorly vascularized and immunologically resistant tumor regions that are largely inaccessible to conventional delivery modalities. | | | |

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| | A core enabling element of the platform is a **proprietary bacteria–payload conjugation system**, which allows stable attachment of cytotoxic payloads to live bacterial carriers. This conjugation system is designed to:<br><br>• Achieve high drug-to-bacteria ratios (DBR ≥ 20), increasing local payload availability<br>• Maintain stability during systemic circulation with payload activity restricted to tumor tissue<br>• Support compatibility with multiple cytotoxic payload classes without re-engineering the bacterial vector<br><br>Within this project, cytotoxic payloads will be designed and optimized for tumor-restricted delivery, including compounds based on established pharmacological frameworks such as MMAE- and DXd-related mechanisms. Payload selection emphasizes extreme potency, induction of immunogenic cell death, and reduced systemic toxicity when delivered locally.<br><br>The therapeutic strategy centers on **combination treatment**, pairing a systemically administered ICI with a tumor-localized BDC. The project will evaluate whether intratumoral cytotoxic delivery enhances antigen release, immune activation, and responsiveness to immune checkpoint blockade, thereby improving outcomes in immunologically resistant solid tumors. |
| **Rationale for Joint R&D Project** | **Rationale for Industry–Research Institute Collaboration**<br><br>This joint R&D project integrates complementary capabilities:<br><br>• The **Korean SME partner** contributes a development-oriented BDC platform optimized for tumor targeting, controlled bacterial persistence, proprietary payload conjugation technology, and regulatory alignment.<br>• The **research institute** contributes expertise in cytotoxic payload design, AI-driven optimization, mechanistic evaluation, and genetic stability analysis.<br><br>This structure enables parallel optimization of payloads and delivery technology, accelerates translational assessment, and supports progression toward IND-enabling studies and downstream licensing |
| **Key Words** | 1. Bacteria Drug Conjugate (BDC)<br>2. Bacteria-Enhanced Linker Cancer Removal Optimization (BELCRO)<br>3. AI-Driven Payload Discovery and Optimization<br>4. Regulatory-Aligned Live Biotherapeutic Product (LBP) Development |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 4 ) | End of the Project | Level ( 6 ) |
|---|---|---|---|---|

**R&D Objectives**

**Overall Objective**

To establish a **clinically translatable, tumor-selective Bacteria–Drug Conjugate platform** capable of delivering next-generation cytotoxic payloads in combination with immune checkpoint inhibitors, and to generate an **IND-ready, licensing-ready preclinical data package**, advancing the technology from **TRL 4 to TRL 6**.

**Specific Objectives**

1. Develop a tumor-targeting BDC platform suitable for combination immunotherapy
2. Establish a high-capacity, modular bacteria–payload conjugation system
3. Discover and optimize cytotoxic payloads using AI-driven approaches
4. Engineer rational combination strategies with patent-expiring ICIs
5. Validate genetic stability and biosafety aligned with FDA LBP guidance
6. Generate a comprehensive IND-ready preclinical dataset

**R&D Target Specifications**

**Tumor Targeting and Biodistribution**

- Tumor-to-normal tissue bacterial ratio: **≥ 1,000:1**
- Intratumoral localization beyond perivascular regions: **≥ 70%** of signal >100 μm from vessels
- Clearance from non-tumor tissues: **≥ 99.9% within 14 days**

**Payload Conjugation Performance**

- Mean DBR: **≥ 20** (CV ≤ 25%)
- Serum stability: **≥ 90% payload retention at 24 h**
- Payload classes conjugated: **≥ 2** distinct classes
- Post-conjugation bacterial viability: **≥ 80%**

**Cytotoxic Payload Activity**

- In vitro potency: **$IC_{50}$ ≤ 10 nM**
- Therapeutic index: **≥ 10-fold vs non-tumor cells**
- Immunogenic cell death: **≥ 2-fold increase in DAMP markers**

**In Vivo Antitumor Efficacy**

- Tumor growth inhibition: **≥ 70%** vs vehicle
- Survival benefit: **≥ 50% increase vs ICI monotherapy**
- Dose-sparing effect: **≥ 30% ICI dose reduction with maintained efficacy**

**Genetic Stability and Biosafety**

- No conjugation cassette loss over **≥ 50 generations**
- No detectable horizontal gene transfer
- No treatment-related mortality or uncontrolled persistence

**IND-Enabling Readiness**

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| | • Batch-to-batch DBR variation: **≤ 30%**<br>• Formulated stability: **≥ 4 weeks**<br>• Data package aligned with FDA LBP guidance |
| **R&D Plan** | **Phase I (Year 1): Platform Establishment**<br><br>• Optimize tumor-targeting bacterial vector<br>• Establish baseline performance of the bacteria–payload conjugation system<br>• Initiate AI-driven payload discovery<br>• Define genetic stability baseline<br><br>**Phase II (Year 2): Payload Optimization and Combination Validation**<br><br>• Optimize payload potency and conjugation compatibility<br>• Conduct in vivo efficacy and ICI combination studies<br>• Expand genetic stability and biosafety validation<br><br>**Phase III (Year 3): IND-Enabling Package and Licensing Readiness**<br><br>• Complete IND-oriented efficacy, biodistribution, and safety studies<br>• Integrate datasets aligned with FDA LBP guidance<br>• Prepare licensing-ready development package |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | The research institute will provide **scientific leadership and translational differentiation**, including:<br><br>• AI-driven discovery and optimization of cytotoxic payloads<br>• Mechanistic studies supporting immunogenic cell death and immune modulation<br>• Genetic stability modeling and NGS-based validation<br>• Contribution to translational and regulatory science strategy |
| **Expected R&R of Korean SME Partner** | The Korean SME partner will serve as the **primary development owner**, including:<br><br>**1.** Engineering of tumor-targeting bacterial strains<br>**2.** Development and characterization of the proprietary bacteria–payload conjugation system<br>**3.** In vitro and in vivo efficacy, safety, and combination studies |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | **Technological Outcomes**<br><br>• Establishment of a **first-in-class Bacteria–Drug Conjugate platform**<br>• Validation of a scalable, modular bacteria–payload conjugation strategy<br>• Integration of AI-driven payload discovery with live bacterial delivery<br><br>**Commercial and Industrial Impact**<br><br>• Delivery of an **IND-ready licensing asset**<br>• Enablement of lifecycle extension strategies for ICIs nearing patent expiration<br>• Creation of a repeatable platform expandable across solid tumor indications<br><br>**Societal and Strategic Impact**<br><br>• Advancement of live biotherapeutics as a credible oncology modality<br>• Improved therapeutic options for immunologically resistant solid tumors<br>• Strengthening of international academia–industry collaboration models |

## 6. References

(1) Forbes NS. Nat Rev Cancer 2010. Engineering the perfect (bacterial) cancer therapy

(2) Chowdhury S et al. Nat Med 2019. Programmable bacteria induce durable tumor regression and systemic antitumor immunity

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(4) Sizuo Y et al. Front. Bioeng Biotechnol 2025. Bacterial carrier-mediated drug delivery systems: a promising strategy in cancer therapy

(5) Besan H. AS et al. Caners 2021. Integration of Salmonella into Combination Cancer Therapy

(6) Dino K et al. Oncotarget 2017. Local application of bacteria improves safety of Salmonella-mediated tumor therapy and retains advantages of systemic infection

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